

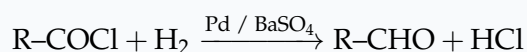
ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

Class 12th Chemistry · Complete Named Reactions & Reaction Mechanisms

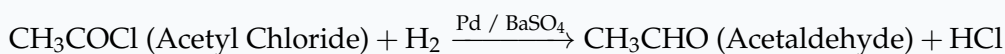
1. Rosenmund Reduction

Preparation of Aldehydes from Acyl Chlorides

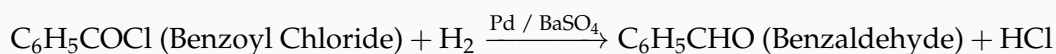
Principle: Catalytic hydrogenation of an acyl chloride (acid chloride) using hydrogen gas in the presence of palladium supported on barium sulfate (Pd/BaSO₄), partially poisoned with sulfur or quinoline.



Specific Example:



Aromatic Example:

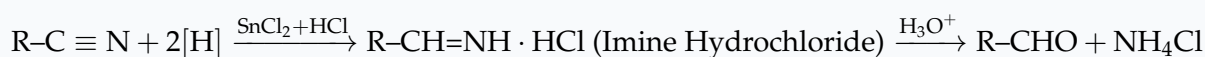


- **Role of BaSO₄:** Acts as a catalyst poison to reduce the catalytic activity of palladium, preventing further reduction of the aldehyde into a primary alcohol.
- **Limitation:** Formaldehyde (HCHO) cannot be prepared by this method because formyl chloride (HCOCl) is unstable at room temperature.

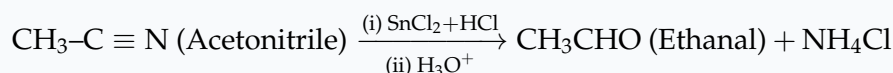
2. Stephen Reaction (Stephen Reduction)

Preparation of Aldehydes from Nitriles

Principle: Nitriles (alkyl or aryl cyanides) are reduced with stannous chloride (SnCl₂) and concentrated hydrochloric acid (HCl) to form an imine intermediate, which upon acidic hydrolysis yields the corresponding aldehyde.



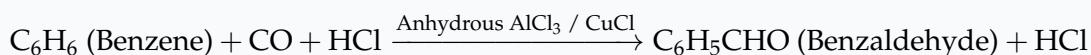
Specific Example:



3. Gattermann-Koch Reaction

Formylation of Benzene to Benzaldehyde

Principle: When benzene or its derivatives are treated with carbon monoxide (CO) and hydrogen chloride (HCl) gas in the presence of anhydrous aluminium chloride (AlCl₃) or cuprous chloride (CuCl), benzaldehyde is produced.



■ **Active Intermediate:** Formyl chloride (HCOCl) is generated *in situ* from CO + HCl, which yields the formyl cation (CHO⁺) as the electrophile in the presence of Lewis acid AlCl₃.

4. Etard Reaction

Controlled Oxidation of Toluene to Benzaldehyde

Principle: Toluene is partially oxidized using chromyl chloride (CrO₂Cl₂) in a non-polar solvent like carbon disulfide (CS₂) or carbon tetrachloride (CCl₄) to form a brown chromium complex, which on subsequent acidic hydrolysis yields benzaldehyde.



■ **Significance:** Provides a selective, controlled oxidation method that prevents the over-oxidation of toluene into benzoic acid.

5. Ozonolysis of Alkenes

Cleavage of Carbon-Carbon Double Bonds

Principle: Alkenes react with ozone (O₃) to form an ozonide intermediate. Subsequent reductive cleavage using zinc dust and water (Zn / H₂O) yields aldehydes, ketones, or a mixture of both, depending on the substitution pattern of the alkene.

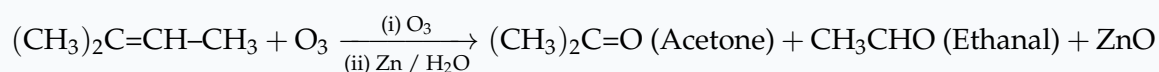


Specific Examples:

1. **Ethene to Formaldehyde:**



2. **2-Methylbut-2-ene to Acetone and Acetaldehyde:**

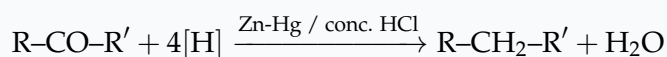
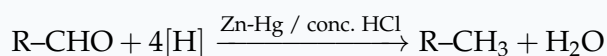


■ **Role of Zinc (Zn):** Acts as a reducing agent that scavenges hydrogen peroxide (H₂O₂) formed during hydrolysis, preventing aldehydes from oxidizing into carboxylic acids.

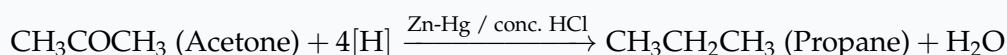
6. Carbonyl-to-Methylene Reductions

A. Clemmensen Reduction (Acidic Medium)

Principle: Aldehydes or ketones are reduced to their corresponding alkanes by treatment with zinc amalgam (Zn-Hg) and concentrated hydrochloric acid (HCl).



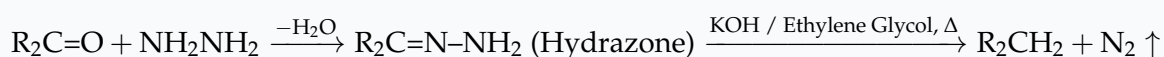
Example:



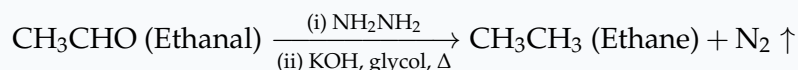
■ **Suitability:** Best suited for acid-stable carbonyl compounds; sensitive to base-labile groups.

B. Wolff-Kishner Reduction (Basic Medium)

Principle: Carbonyl compounds are treated with hydrazine (NH_2NH_2) to form a hydrazone, which upon heating with a strong base (potassium hydroxide, KOH) in a high-boiling solvent like ethylene glycol ($\text{HO-CH}_2\text{CH}_2\text{-OH}$) yields an alkane with nitrogen gas release.



Example:



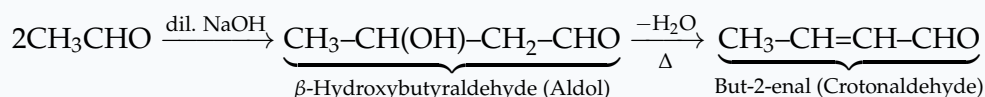
■ **Suitability:** Ideal for base-stable, acid-sensitive substrates.

7. Aldol Condensation & Cross-Aldol Condensation

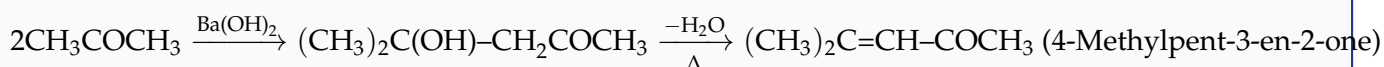
Reactions of Carbonyls with α -Hydrogen

Requirement: The aldehyde or ketone must possess at least one α -hydrogen atom.

Principle: In the presence of a dilute alkali (dil. NaOH or Ba(OH)_2), two molecules of an aldehyde or ketone containing α -hydrogen atoms undergo nucleophilic addition to yield a β -hydroxyaldehyde (aldol) or β -hydroxyketone (ketol). Upon heating, dehydration occurs to form an α, β -unsaturated carbonyl compound.



Condensation of Ketones:



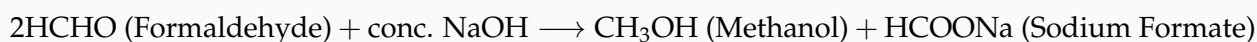
■ **Cross-Aldol Condensation:** When aldol condensation is conducted between two distinct carbonyl compounds, a mixture of up to four different products is obtained if both possess α -hydrogens.

8. Cannizzaro Reaction

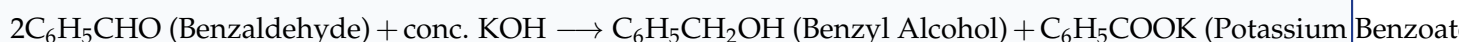
Disproportionation of Carbonyls lacking α -Hydrogen

Requirement: The aldehyde must lack any α -hydrogen atoms (e.g., HCHO , $\text{C}_6\text{H}_5\text{CHO}$, $(\text{CH}_3)_3\text{C-CHO}$).

Principle: In the presence of concentrated alkali (50% NaOH or KOH), aldehydes with no α -hydrogen undergo self-oxidation and reduction (disproportionation / redox reaction). One molecule is reduced to an alcohol, while the other is oxidized to a carboxylic acid salt.



Aromatic Example:

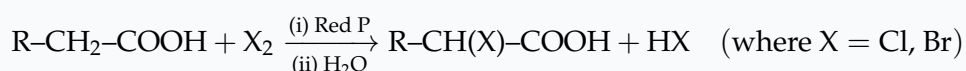


9. Hell-Volhard-Zelinsky (HVZ) Reaction

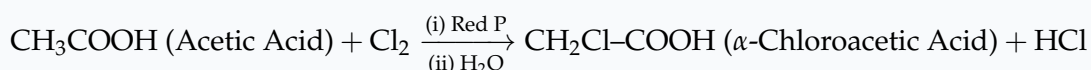
α -Halogenation of Carboxylic Acids

Requirement: Carboxylic acids possessing at least one α -hydrogen atom.

Principle: Aliphatic carboxylic acids react with chlorine (Cl_2) or bromine (Br_2) in the presence of a catalytic amount of red phosphorus (red P) to selectively replace α -hydrogen atoms with halogen atoms, producing α -halocarboxylic acids.



Specific Example:



■ Formic acid (HCOOH) does not undergo the HVZ reaction because it lacks an α -carbon and α -hydrogen.

10. Comparative Summary Table

Reaction	Starting Material	Reagents	Product Formed
Rosenmund	Acyl Chloride (RCOCl)	H_2 , Pd/BaSO_4	Aldehyde (RCHO)
Stephen	Nitrile (R-CN)	$\text{SnCl}_2 + \text{HCl}$, then H_3O^+	Aldehyde (RCHO)
Gattermann-Koch	Benzene (C_6H_6)	$\text{CO} + \text{HCl}$, anhyd. AlCl_3	Benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$)
Etard	Toluene ($\text{C}_6\text{H}_5\text{CH}_3$)	$\text{CrO}_2\text{Cl}_2/\text{CS}_2$, then H_3O^+	Benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$)
Ozonolysis	Alkene ($\text{R}_2\text{C=CR}'_2$)	O_3 , then $\text{Zn/H}_2\text{O}$	Carbonyl compounds ($\text{CHO} / \text{C=O}$)
Clemmensen	Carbonyl (C=O)	$\text{Zn-Hg} + \text{conc. HCl}$	Alkane ($-\text{CH}_2-$)
Wolff-Kishner	Carbonyl (C=O)	NH_2NH_2 , then $\text{KOH/glycol}, \Delta$	Alkane ($-\text{CH}_2-$)
Aldol	Carbonyl with α -H	dil. NaOH, Δ	α, β -Unsaturated Carbonyl
Cannizzaro	Aldehyde without α -H	conc. NaOH (50%)	Alcohol + Carboxylic Acid Salt
HVZ	Carboxylic acid with α -H	$\text{X}_2/\text{Red P}$, then H_2O	α -Halocarboxylic acid